厦门大学《微积分 I-2》课程期中试卷

试卷类型：（理工类 A 卷） 考试日期

一、填空题：

1. -2　　2. (1).$z=x^2+y^2,\ (1\le z\le 4)$　(2).$\begin{cases}1\le x^2+y^2\le 4\\ z=0\end{cases}$

3. $y=xe^x+C_1e^x+C_2e^{2x}$　　4. $k=2,\ P=2$　　5. $dz_{|(0,0)}=\frac{1}{3}dx+\frac{1}{3}dy$

6. $\frac{7\sqrt{6}}{6}$

二、 所求平面方程为$x-y=0.$

(思路：关键：可设过$L:x=y=2z\Leftrightarrow\begin{cases}x-y=0\\ y-2z=0\end{cases}$的

平面π：$\lambda(x-y)+\mu(y-2z)=0$

三、1. $y=\frac{1}{2C}-\frac{1}{2}Cx^2.$

(思路：原微分方程变形为$y'=-\sqrt{1+\left(\frac{y}{x}\right)^2}+\frac{y}{x}$

令$\frac{y}{x}=u\Rightarrow y=xu$　　$\therefore\frac{dy}{dx}=u+x\frac{du}{dx}$

2. $y=(x+1)^4-1.$

(思路：$i)$令$y'=P(x),$则$y''=\frac{dP(x)}{dx},\therefore$原微分方程变形为

$$\frac{dP}{dx}-\frac{1}{1+x}P=8(1+x)^2\cdots\cdots(1)$$

1)对应齐次方程为：$\frac{dP}{dx}-\frac{1}{1+x}P=0$解得：$P=C(1+x)$

2)(常数变易法)可设(1)的通解为$P=u(x)(1+x).$

代入解得$u(x)=4(1+x)^2+C_1.$

$\therefore$可设(1)的通解为$P=4(1+x)^3+C_1(1+x)$

3)由初始条件$P(0)=y'(0)=0$得:$C_1=0$

$\therefore$可设(1)的特解为$P=4(1+x)^3$

$ii)\ y'=4(1+x)^3\Rightarrow y=(1+x)^4+C_2$

四、1. $L:\begin{cases}x=1+\cos\theta\\ y=\sin\theta\\ z=2\sin\dfrac{\theta}{2}\end{cases}$ (θ为参数). 2. 所求切线: $\dfrac{x-1}{-1}=\dfrac{y-1}{0}=\dfrac{z-\sqrt{2}}{\frac{\sqrt{2}}{2}}$

(思路: 关键: 切线L的方向$\vec{v}=\left(\dfrac{dx}{d\theta},\dfrac{dy}{d\theta},\dfrac{dz}{d\theta}\right)\Big|_{P_0}=\left(-\sin\theta,\cos\theta,\cos\dfrac{\theta}{2}\right)\Big|_{P_0}$

五、

$$1.\frac{\partial u}{\partial t}=-\frac{3}{2t}\cdot\frac{1}{(\sqrt{4\pi t})^3}e^{-\frac{x^2+y^2+z^2}{4t}}+\frac{1}{(\sqrt{4\pi t})^3}e^{-\frac{x^2+y^2+z^2}{4t}}\cdot\frac{x^2+y^2+z^2}{4t^2}$$

$$=\left(-\frac{3}{2t}+\frac{x^2+y^2+z^2}{4t^2}\right)\frac{1}{(\sqrt{4\pi t})^3}e^{-\frac{x^2+y^2+z^2}{4t}}$$

$$2.i)\frac{\partial u}{\partial x}=\frac{1}{(\sqrt{4\pi t})^3}e^{-\frac{x^2+y^2+z^2}{4t}}\cdot\left(-\frac{x}{2t}\right)\qquad ii).\frac{\partial^2 u}{\partial x^2}=\left(-\frac{1}{2t}+\frac{x^2}{4t^2}\right)\frac{1}{(\sqrt{4\pi t})^3}e^{-\frac{x^2+y^2+z^2}{4t}}$$

3. 同理:$\dfrac{\partial^2 u}{\partial y^2}=\left(-\dfrac{1}{2t}+\dfrac{y^2}{4t^2}\right)\dfrac{1}{(\sqrt{4\pi t})^3}e^{-\frac{x^2+y^2+z^2}{4t}}$, $\dfrac{\partial^2 u}{\partial z^2}=\left(-\dfrac{1}{2t}+\dfrac{z^2}{4t^2}\right)\dfrac{1}{(\sqrt{4\pi t})^3}e^{-\frac{x^2+y^2+z^2}{4t}}$

六、1. 连续

(思路: $\lim\limits_{(x,y)\to(0,0)}f(x,y)=\lim\limits_{(x,y)\to(0,0)}\dfrac{x\left(x^2-y^2\right)}{x^2+y^2}$

令:$\begin{cases}x=\rho\cos\theta\\ y=\rho\sin\theta\end{cases}$

$$\therefore\lim_{(x,y)\to(0,0)}f(x,y)=\lim_{\rho\to0}\frac{\rho\cos\theta\cdot\rho^2\cos2\theta}{\rho^2}=0$$

2. 偏导数存在

(思路: $i)(x,y)=(0,0)$,

$$f_x'(0,0)=\lim_{\Delta x\to0}\frac{f(0+\Delta x,0)-f(0,0)}{\Delta x}=\cdots=1$$

$$f_y'(0,0)=\lim_{\Delta y\to0}\frac{f(0,0+\Delta y)-f(0,0)}{\Delta y}=\cdots=0$$

$$(ii)(x,y)\neq(0,0),\quad \because f(x,y)=\frac{x\left(x^2-y^2\right)}{x^2+y^2}.$$

$$\therefore f_x'(x,y)=\frac{\left(3x^2-y^2\right)\cdot\left(x^2+y^2\right)-\left(x^3-xy^2\right)\cdot 2x}{\left(x^2+y^2\right)^2}$$

$$f_y'(x,y)=\frac{(-2xy)\cdot\left(x^2+y^2\right)-\left(x^3-xy^2\right)\cdot 2y}{\left(x^2+y^2\right)^2}$$

3. 不可微

(思路：ii)$\Delta z=f(0+\Delta x,0+\Delta y)-f(0,0)=\dfrac{\Delta x\cdot\left[(\Delta x)^2-(\Delta y)^2\right]}{(\Delta x)^2+(\Delta y)^2}$

ii)求 $\lim\limits_{(\Delta x,\Delta y)\to(0,0)}\dfrac{\Delta z-f_x'(0,0)\Delta x-f_y'(0,0)\Delta y}{\rho}=\cdots=-\dfrac{1}{\sqrt{2}}\neq 0$

$\therefore \Delta z-f_x'(0,0)\Delta x-f_y'(0,0)\Delta y\neq o(\rho)$

$\therefore f(x,y)$在$(0,0)$不可微

七、极大值点$(0,-1)$,极大值为-1;　极小值点$(0,-1)$,极小值为-1;

（思路：$i)\because x^2+y^2-2yz-z^2+2=0\quad\therefore 2xdx+2ydy-2(z\,dy+ydz)-2zdz=0$

$$\therefore dz=\frac{x}{y+z}dx+\frac{y-z}{y+z}dy,\quad \frac{\partial z}{\partial x}=\frac{x}{y+z},\frac{\partial z}{\partial y}=\frac{y-z}{y+z}$$

$$ii)\begin{cases}\dfrac{\partial z}{\partial x}=\dfrac{x}{y+z}=0\\ \dfrac{\partial z}{\partial y}=\dfrac{y-z}{y+z}=0\end{cases}\Rightarrow\begin{cases}x=0\\ y=z\end{cases}\cdots\cdots(1)$$

$\because$又有：$x^2+y^2-2yz-z^2+2=0\cdots\cdots(2)$

(1)(2)得：$\begin{cases}x=0\\ y=z=1\end{cases}$或$\begin{cases}x=0\\ y=z=-1\end{cases}$

iii)求$A=\dfrac{\partial^2 z}{\partial x^2},C=\dfrac{\partial^2 z}{\partial y^2},B=\dfrac{\partial^2 z}{\partial x\partial y}$

$$1)A=\frac{\partial^2 z}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial}{\partial x}\left(\frac{x}{y+z}\right)=\frac{(y+z)-x\cdot\dfrac{\partial z}{\partial x}}{(y+z)^2}$$

$$C=\frac{\partial^2 z}{\partial y^2}=\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right)=\frac{\partial}{\partial y}\left(\frac{y-z}{y+z}\right)=\frac{\left(1-\dfrac{\partial z}{\partial y}\right)(y+z)-(y-z)\cdot\left(1+\dfrac{\partial z}{\partial y}\right)}{(y+z)^2}$$

$$B=\frac{\partial^2 z}{\partial x\partial y}=\frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)=\frac{\partial}{\partial y}\left(\frac{x}{y+z}\right)=\frac{-x\cdot\left(1+\frac{\partial z}{\partial y}\right)}{(y+z)^2}$$

2)点$(0,1,1)$, $AC-B^2=\frac{1}{2}\times\frac{1}{2}-0^2>0$, 且$A>0$

点$(0,-1,-1)$, $AC-B^2=\left(-\frac{1}{2}\right)\times\left(-\frac{1}{2}\right)-0^2>0$, 且$A<0$

八、 $f''_{11}(x,2x)=-2x^2$